YASSER AHMAD RAAFAT

Senior AI Engineer · Systems Architect · Full-Stack Engineer

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EDUCATION

Bachelor of Computer Engineering **2020 – 2025**

Modern Academy for Engineering and Technology *Cairo, Egypt*

TECHNICAL SKILLS

AI & LLM Engineering: HuggingFace Transformers, PyTorch, LoRA / QLoRA / PEFT, RLHF, DPO, SFT, Reward Modeling, LLM Alignment, Synthetic Data Generation, Prompt Engineering, Multi-agent Orchestration, Agentic Runtimes, Tool Use / Function Calling, MCP (Model Context Protocol), Claude Code, Anthropic SDK, OpenRouter

Agentic Systems & RAG: LangChain, LangGraph, LlamaIndex, Hybrid RAG (dense + sparse + reranking), pgvector, Inverted-Index + Trigram / Edit-Distance Search, RRF Fusion, Vector Embeddings, Context & Memory Architecture, Agent-Authored MCP Servers

Languages & Scripting: Python (async / sync), JavaScript (ES6+), TypeScript, SQL, MySQL, Bash / Shell Scripting, Java, C, C++

Backend & API Design: Django, FastAPI, Node.js, Express, REST API Design, GraphQL, Microservices Architecture, Event-driven Architecture, WebSockets, SSE, OAuth2 / JWT, RBAC & ReBAC, Rate Limiting, Concurrency & Async Programming

Data & Databases: PostgreSQL (+ pgvector), MySQL, Redis (caching / pub-sub / streams), MongoDB, SQLAlchemy, Alembic, Pandas, NumPy, Query Optimization, Database Indexing, Migrations

Infrastructure & DevOps: Linux, Docker (sandbox orchestration), Kubernetes, AWS (EC2, S3, CloudWatch), Nginx, GitHub Actions CI/CD, Render, Playwright, Pytest, Vitest, Postman

Frontend: React, Vite, Zustand, Tailwind CSS, Socket.io, SSE-driven UI, Responsive / Theme-aware Design

Model Evaluation & Post-Training: Benchmark Design (SWE-bench Verified, HumanEval, SEAL), Preference Data Pipelines, Ragas, LangSmith, MLflow, Weights & Biases, Red-teaming, Reward Signal Engineering

PROFESSIONAL EXPERIENCE

Founder & Chief Architect **Jan 2023 – Present**

CoreFusion — Agentic Developer Platform | Independent, Live Production Product

- Architected and shipped **CoreFusion**: import any Git repository and an autonomous LLM agent reads, edits, runs, and ships it inside **isolated Docker sandboxes** — wrapped in a marketplace, LMS, and full payments ecosystem so work can be shared, sold, taught, and delegated.
- Codified a **5-rule architecture constitution** enforcing a hard sovereign/executor split: the agent owns every decision (what to read, what to run, when a build is healthy, how to retry); the backend is a stateless primitive executor that provides only container lifecycle, filesystem I/O, memory storage, and LLM transport — and names no language, framework, build tool, or file path.
- Built the **sovereign agent runtime** and session-scoped **Docker sandbox orchestration**: byte-identical cached system prefix, agent-owned append-only context store, per-conversation file versioning, Pre/PostToolUse hook registry, capability-gated tool dispatch, and provider-agnostic LLM transport (OpenRouter) streamed end-to-end over **SSE** with zero polling — plus snapshot, revert, exec, and log-streaming with semantic container kinds and live app previews.
- Designed **Book-RAG**, a three-phase code-intelligence pipeline: repository to navigable "books," workflow chains, cross-workflow links, and keyword-to-symbol summaries — with a **grep-vs-RAG decision layer** so each turn loads only the context it needs without over-fetching.
- Built **MIP**, an RBAC-scoped memory and retrieval plane on **PostgreSQL + pgvector**: hybrid dense/sparse search, trigram and edit-distance typo tolerance, **RRF-fused ranking**, a six-check write gate, soft-delete lifecycle, six memory scopes, and auto-generated user personas for turn-one context grounding.
- Created a **previewable-in-chat action spine** — one propose → human-Accept → apply seam reused across recipe runs, LMS publishing, MCP wiring, and provisioning — guaranteeing no side effect ships without an explicit human gate.

- Implemented a **runtime MCP integration layer**: multi-transport (stdio, HTTP, sandboxed, native), ReBAC dispatch gate, agent-authored server pipeline (draft → validate → sandbox-test → publish), marketplace auto-discovery and one-click autowiring, and a **credential vault the model can never read**.
- Shipped a **commerce and creator ecosystem** on a single payments/ledger/entitlements spine: marketplace, per-user wallets/payouts, bounty escrow, and a full **LMS** (generate → author → publish → sell → certify).
- Established **engineering-integrity infrastructure**: a self-describing repository with per-directory structure maps and a flow router, on-disk plan/status ledgers, a wedge-proof **Playwright** verification harness, and a CI gate of ∼**53 Python suites + full Vitest** under a hard "not done until proven live end-to-end" rule.

AI Engineer — Frontier Model Evaluation & Post-Training **May 2026 – Present**

Alignerr | Remote

- Led dedicated post-training projects targeting **Claude Code** fine-tuning: elevated multi-step reasoning quality, tool-use reliability, and repository-level context handling in production agentic coding systems.
- Designed coding benchmarks and expert preference feedback loops that converted senior engineering judgment into high-signal **RLHF, DPO, and SFT** training data for frontier models.
- Served as a precision evaluator within a select engineering cohort focused on model behavior verification, reasoning chain quality, and production-grade evaluation protocols for frontier labs.

LLM Post-Training Lead **Sep 2024 – Apr 2026**

Outlier AI (Scale AI Platform) | Remote

- Led a distributed network of **2,000+ expert contributors** running large-scale **RLHF, DPO, and SFT** campaigns across major frontier models: GPT-4o/4.5, Claude 3.5/4, Gemini 2.x, Llama 3/4.
- Drove annotation and data curation for **Cursor AI's fast-apply Llama-3-70B fine-tune**: model **surpassed GPT-4 and GPT-4o** on full-file edit accuracy with a ∼**13x inference speedup** exceeding **1,000 tokens/sec** (450-edit benchmark graded by Claude-3-Opus).
- Generated **40,000+ multilingual preference prompts** for Perplexity AI's r1-1776 (DeepSeek-R1): achieved a **100% uncensored response rate** vs. an 85% baseline while preserving scores across MMLU, DROP, MATH-500, and AIME.
- Directed **SWE-bench Verified and SEAL** benchmark campaigns: SFT datasets raised task accuracy **40%**, Python automation cut processing time **40%**, and QA frameworks reduced annotation defects **35%**.

Full-Stack Software Engineer **Dec 2025 – Mar 2026**

Clove Restaurant Maadi | Cairo, Egypt

- Architected and shipped a full production restaurant platform (**React + Zustand + Tailwind**, **Node.js/Express**, **PostgreSQL + MySQL**) on Render with GitHub Actions CI/CD.
- Built **real-time WebSocket chat** (Socket.io, fully DB-persisted), a **gamified loyalty engine** (milestones, avatars, cashback wallet), an admin dashboard, live GPS tracking, and concurrent-safe operations with query-optimized database access.

Software Engineering Instructor **Jun 2024 – Aug 2024**

iSchool — EGY FWD Initiative (DEMI) | Cairo, Egypt

- Mentored **50+ students** in Python and JavaScript; built an automated grading system that raised instructional efficiency **30%** and drove a **95%** project completion rate.

CERTIFICATIONS & ACHIEVEMENTS

Redis University (2024): Certified Developer — RU101, RU102PY, RU202, RU203, RU204, RU301, RU330

freeCodeCamp: Responsive Web Design | JavaScript Algorithms & Data Structures | Back End Development & APIs | Relational Database

HackerRank: SQL (Basic / Intermediate / Advanced) | Software Engineer | Python | **ICPC 2022 — Participant**